Sankalp Vaish

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GitHub | Linkedin | Google Scholar | Portfolio

PROFESSIONAL SUMMARY

I am a Computer Science PhD candidate at UMass Boston with a strong focus on **multimodal AI, computer vision, machine learning, and intelligent automation**. Beyond research, I am passionate about teaching and mentorship, with experience teaching supporting the next generation of computing professionals. I also build and deploy applied machine learning systems in production, including RAGs, LLM-powered applications, and intelligent workflow automation. Passionate about leveraging AI to solve challenging problems in **healthcare, customer intelligence, automation, and biometric authentication**.

TECHNICAL SKILLS

Programming Languages & Scripts	:	PYTHON, JAVA, GNU Octave, C, C++, Solidity, JavaScript, TypeScript.
Core AI / ML & Research Areas	:	Transformers, LLMs, RAGs, Multimodal Learning, Machine learning, Computer Vision.
Database and Frameworks	:	MongoDB, MySQL, PostgreSQL, SQL Server, ChromaDB, Qdrant, Odoo, FastAPI.
Web Frameworks & Tools	:	React, Django, FastAPI, Node.js, Express.js, Vite, HTML5, CSS, jQuery, THREE.js.
Cloud Platforms and CI/CD tools	:	Amazon Web Services, Microsoft Azure, GitHub and GitLab, Web APIs, Bedrock, Neon.

ACADEMIC PROFILE

University of Massachusetts Boston — Sep '24 – May '27
PhD: Computer Science, GPA: 4.0

University of Massachusetts Boston — Sep '22 - May '24
Master of Science: Computer Science, GPA: 4.0

National Institute of Technology, Warangal — Sep '21- Aug '22
P.G. Diploma in Machine Learning and Artificial Intelligence, GPA: 4.0

University of Lucknow, Lucknow — Jul '18 - Jun '21
Bachelor of Science: Computer Science and Statistics, GPA: 3.5

RESEARCH PAPERS AND PUBLICATIONS

- **Sankalp Vaish**, Rishank Singh, Sumukh Veeramalla, Caroline Summerour, John Batsis, Xiaohui Liang, **Task-Centric Multimodal Learning for Cognitive Impairment Detection**, IEEE Global Communications Conference: Selected Areas in Communications (SAC): E-Health, **GLOBECOM** 2026. [arXiv]
- **Sankalp Vaish**, Nana Lin, Umang Shah, Rishank Singh, Xiaohui Liang, Marc Pomplun, **An Enhanced Facial Authentication Approach: Optimized Preprocessing and Automated Filtering for Secure and Reliable Authentication**, IEEE Global Communications Conference: Communication & Information System Security, **GLOBECOM** 2026.
- **Sankalp Vaish**, Sumukh Veeramalla, Rishank Singh, Caroline Summerour, John Batsis, Xiaohui Liang, **CogDrift-4D: Hierarchical Multimodal Modeling of Temporal Behavioral Decay for MCI Detection**, IEEE International Conference on E-health Networking, Application & Services, **Healthcom** 2026. [arXiv]
- Rishank Singh, Youxiang Zhu, **Sankalp Vaish**, Caroline Summerour, Xiaohui Liang, John Batsis, **CDBench: Benchmarking Large Language Models for Conversation-Based Dementia Detection**, IEEE Global Communications Conference: Selected Areas in Communications (SAC): E-Health, **GLOBECOM** 2026.
- Rishank Singh, **Sankalp Vaish**, Xiaohui Liang, John Batsis, Jennifer Grace Poole, Danae Gross, **Clinical Geriatric Functional Assessment using Wearable Sensing and Machine Learning**, IEEE International Conference on E-health Networking, Application & Services, **Healthcom** 2024. [DOI]

PROFESSIONAL EXPERIENCE

AI/Machine Learning Internship— Ottometric, Boston — June '26 – Dec '26

- Designed and built an Agentic AI sales assistant with dual LLM integrations (Odoo-native + AWS Bedrock Claude): featuring multi-turn tool-use reasoning, function-calling tools, human-in-the-loop action execution, and F.O.R.E.C.A.S.T. deal scoring.
- Developed AI-powered CRM capabilities within Odoo, enabling automatic deal summaries, meeting note processing, actionable insights, and natural language querying of sales opportunities using LLMs.
- Integrated Amazon Bedrock into Odoo to provide secure access to multiple LLMs for AI-driven sales automation.
- Designed and implemented Gmail and Microsoft Outlook integrations with email enabling, like detection of email opens, link clicks, and replies to provide real-time customer interaction insights within Odoo CRM.
- Built an AI-powered customer simulator leveraging retrieval-augmented generation (RAG) with a vector database to retrieve relevant knowledge from emails, meeting notes, and customer interactions, enabling context-aware customer conversation.
- Built an automated sales funnel solution that streamlines the import of conference attendee data from Excel into Odoo CRM, assigns sales opportunities, and provides analytics on lead generation, pipeline progression, and event conversion rates.
- Restoring and reconciling data from multiple available sources, including Langfuse, Qdrant, WAL backups, HubSpot exports, and the development env., while also developing an Odoo module to identify and restore data from hash-based filestore files.
- Developed public-facing site pages including the privacy policy, user guides and careers pages.

Research Assistant—MobCP Lab, UMass Boston May '25 – Present

- Designed and implemented a multimodal framework combining video (facial Action Units), audio (prosodic cues), and text (linguistic features) for early dementia and depression detection. Using models like MiVOLO, ResEmoteNet and OpenFace.
- Built a voice-based sentiment analysis model that can analyse and predict distress levels from real audio and conversation transcripts from patients dealing with chronic disease, helping improve patient–doctor communication.
- **In collaboration with University of North Carolina:** Conducted comparative experiments across modalities and models (NN vs. LLM) to evaluate detection performance. Built a contactless system for assessing cognitive decline in older adults, analysing facial muscle activity and speech during structured Alexa interactions using multimodal features.
- **In collaboration with Town of Weymouth:** Design and implement a secure federated authentication framework integrating ADFS with Exchange Server 2019 and other enterprise applications. Configured claims-based authentication, custom SSL certificates, and certificate trust using YubiKey-based multi-factor authentication for Windows logins, and Fortinet VPN.

Teaching Fellow—Computer Science, UMass Boston Sep '24 – May '27

Instructor | Data Structures in Java

- Taught Data Structures in Java, covering topics like arrays, lists, stacks, queues, trees, and algorithmic problem-solving.

Instructor | Introduction to Programming in Python

- Taught Introduction to Python, focusing on fundamental programming concepts, data structures, and real-world applications.

Responsibilities- Develop and assess assignments, projects, and exams, Design interactive learning materials for student engagement. Provide mentorship and academic support to enhance problem-solving and coding skills.

Research Assistant—Visual Attention Lab, UMass Boston May '23 – May '25

- Worked on improving image quality by developing various filters for detecting blur or contrast for better face recognition.
- Built a frame annotation tool for labelling objects in video, producing the training datasets to improve object detection models.
- Developed a model to identify and select best image from a set, ensuring optimal data for training and preventing model corruption.
- Implemented advanced techniques to fine-tune YOLOv8 for enhanced object and human detection in security camera footage.
- Performing Knowledge Distillation: transferring knowledge from teacher model like Grounding Dino to student model.
- **In collaboration with MobCP Lab:** developed projects related to healthcare technology, including the detection of: dementia using voice assistant systems and for older adults and physical function such as gait speed, sit-to-stand ratio and balance.
- **In collaboration with Blub0x:** Developed a website using MERN stack to collect data of different users and then used this data to make a classifier to perform this action on its own. Built a real-time system that detects elevator regions, available floor space, current floor, direction of movement and door status from video, for intelligent building monitoring.

Assistant Teacher—Sky Public School, Lucknow, India Dec '21 – Jul '22

- Teaching Python, basic concepts of database and assembly language to junior high school and high school students.
- Responsibilities- Teaching, In-charge of software and hardware solutions.

Professional Service

- **Technical Program Committee (TPC) Member**, IEEE HealthCom 2026 (Main Track).
- **Reviewer**, WiMob 2026 – Trust, Security and Privacy in Wireless and Mobile Networks (SWMN) Symposium.
- **Reviewer**, WiMob 2026 – Next Generation Mobile Networking (NGMN) Symposium.
- **Reviewer**, IEEE GLOBECOM 2026 – Communications and Information System Security (CISS) Symposium.
- **Reviewer**, IEEE International Conference on Metaverse Computing, Networking and Applications (MetaCom 2025).

AWARDS AND ACHIEVEMENTS

- Received Public Choice Award for AI Hackathon hosted by UMASS Boston for Diabetes Distress project, October 31, 2025
- Received grade A in Blockchain Professional Certificate by Edureka, November 17, 2021
- 100% in HTML, CSS, JavaScript certificate with by Johns Hopkins University through Coursera, June 26, 2021
- 91.2% in Applied Machine Learning certificate from University of Michigan through Coursera, August 29, 2020
- 96.16% in Applied Text Mining certificate from University of Michigan through Coursera, Sept 25, 2020
- 96.58% in Introduction to Data Science certificate from University of Michigan through Coursera, Aug 17, 2020
- Secured 91.6% in Neural Networks and Deep Learning certificate through Coursera, August 12, 2020
- Received 96.16% marks in What is Data Science certificate by IBM through Coursera, July 20, 2020
- Achieved 100% marks in Introduction to AI certificate by IBM through Coursera, June 12, 2021
- Awarded for obtaining 100% marks in ICSE Board Examinations in Computer Science, 2016
- Won a gold medal for 14th National Cyber Olympiad, 2015

COURSEWORK AND CERTIFICATIONS

- Earned a certification in Object-Oriented Programming in Java from LinkedIn Learning, September 15, 2022.
- Earned a certification in Data Structures in Java from LinkedIn Learning, September 12, 2022.
- Completed a certification in Deep Learning, Machine Learning, Python, and Pandas through Kaggle, Aug 2020.

INTERESTS

Teaching, Puzzle Solving, Travelling, Reading, Swimming.